

GEN3C: 3D-Informed World-Consistent Video Generation with Precise Camera Control

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Overview

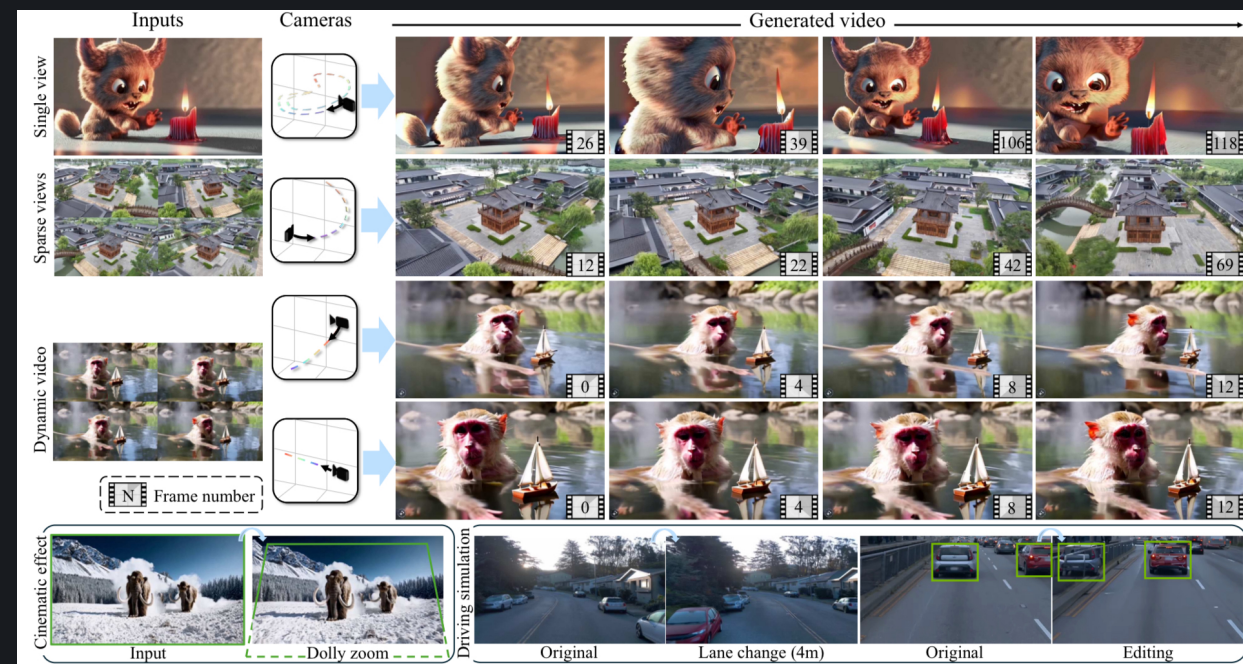
Paper info

CVPR 2025 Highlight

Key words: **Video Diffusion, 3D Consistency, Camera Control**

Authors: **Ren, Xuanchi, et al.**

Institution: **NVIDIA, University of Toronto**

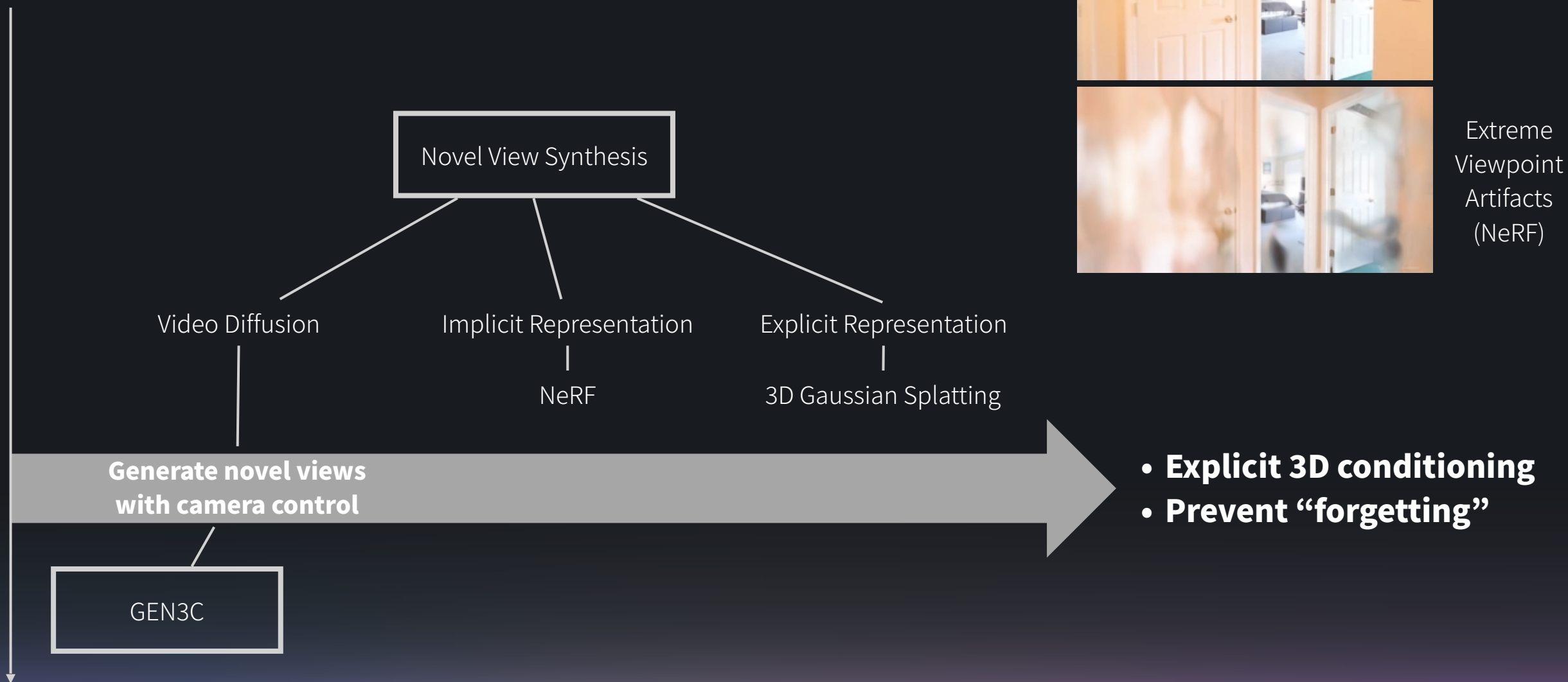


The key issue that the above paper focuses on

3D Consistency, Camera Control →

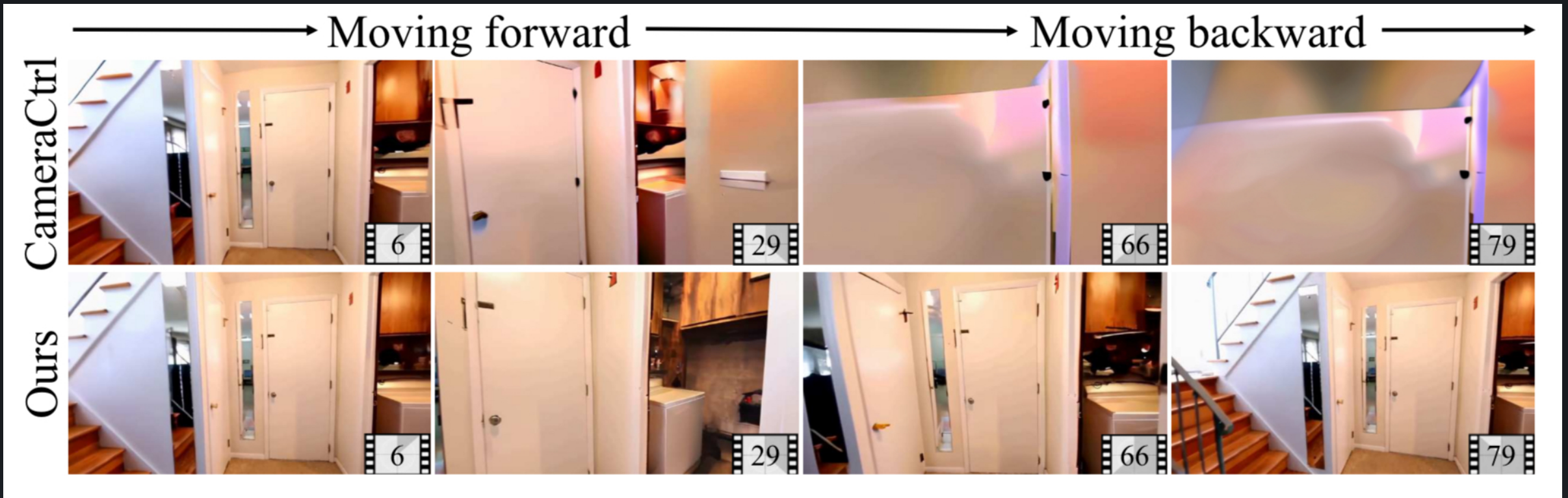
- Building a Spatiotemporal 3D Cache
- Conditioning by Rendering
- Permutation-Invariant Fusion (Max-Pooling In-Layers)

Positioning



Motivation

Existing works like CameraCtrl or MotionCtrl inject camera pose data as numerical parameters



Without explicit 3D guidance,
video diffusion model “forgets” the history



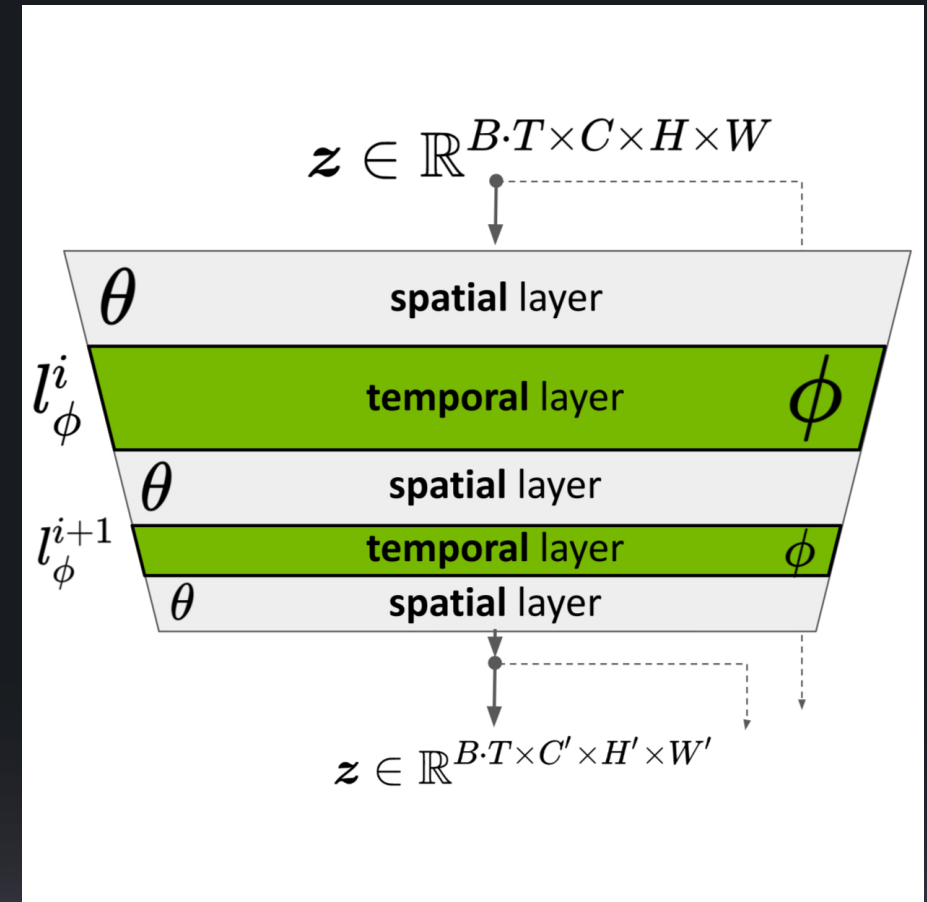
3D Cache

Preliminary **Latent Video Diffusion Model (VDM)**

$$\mathbb{E}_{x_0 \sim p_{data}(x), \tau \sim p_{\tau}, \epsilon \sim \mathcal{N}(0, I)} \left[\left\| f_{\theta}(\underset{\substack{\uparrow \\ \text{Video Sequences} \\ L \times 4 \times \frac{H}{8} \times \frac{W}{8}}}{x_{\tau}}; \underset{\substack{\downarrow \\ \text{3D Cache}}}{c}, \tau) - y \right\|_2^2 \right]$$

L is the number of frames

Pre-trained video diffusion models:
**already “learned” the underlying laws of
 physics, lighting, and temporal continuity**



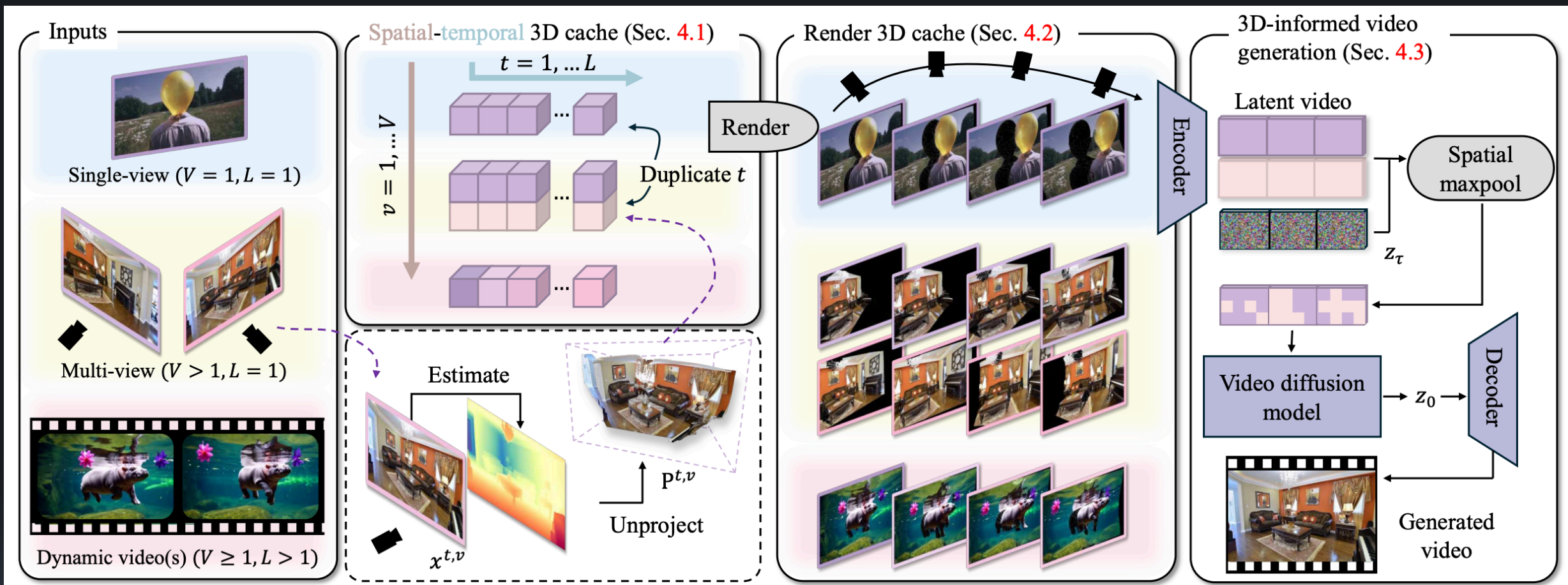
i : layer index

Results

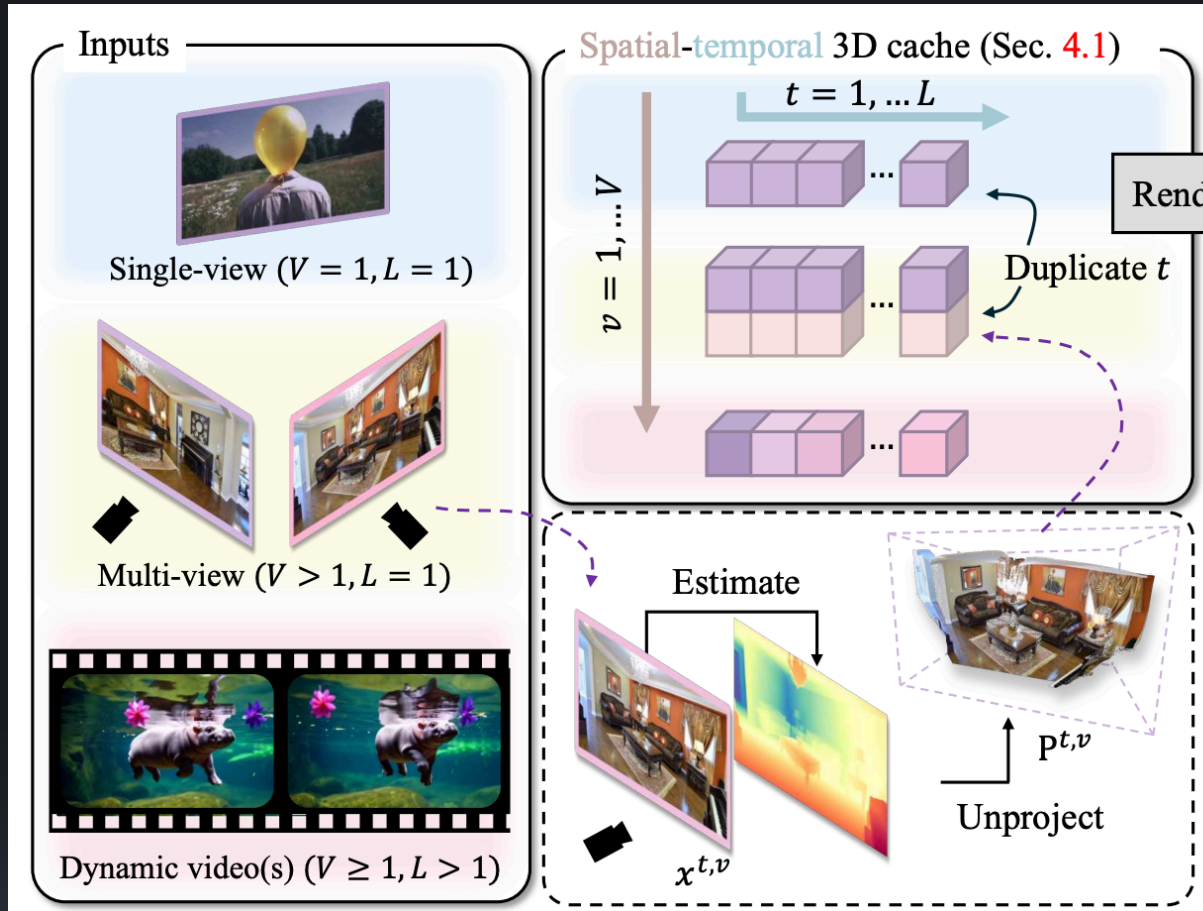


Given a **single** image, GEN3C enables users to create long, consistent videos by specifying a camera trajectory.

Overview



Method Building a Spatiotemporal 3D Cache



- RGB image(s) or pre-generated video frames
- Depth estimation values of input image(s)/frames
- Camera poses of input image(s)/frames

For unprojection

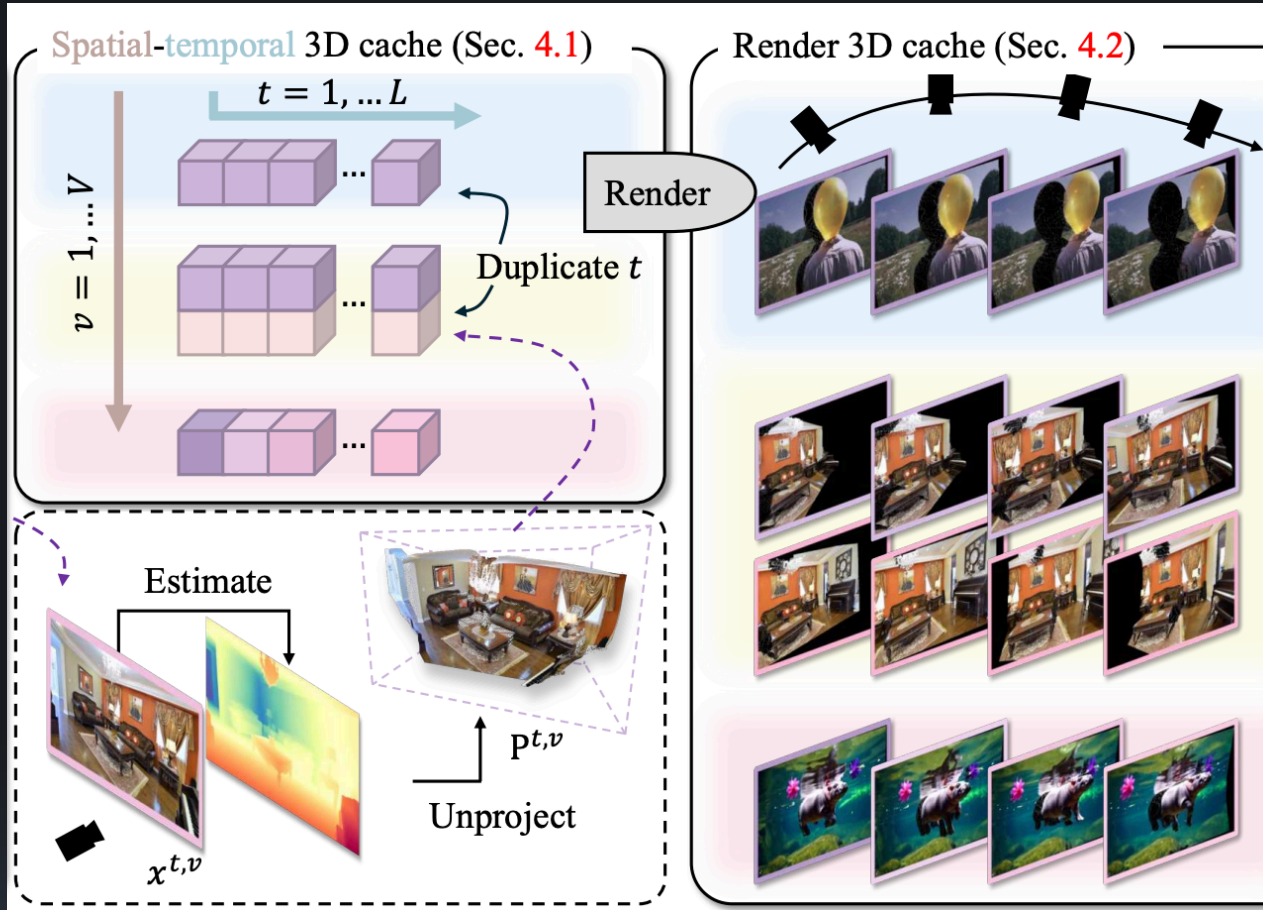
Input

A spatiotemporal 3D cache

This cache acts as the **explicit geometry** used to guide the video generation process.

Output

Method **Rendering the 3D Cache**



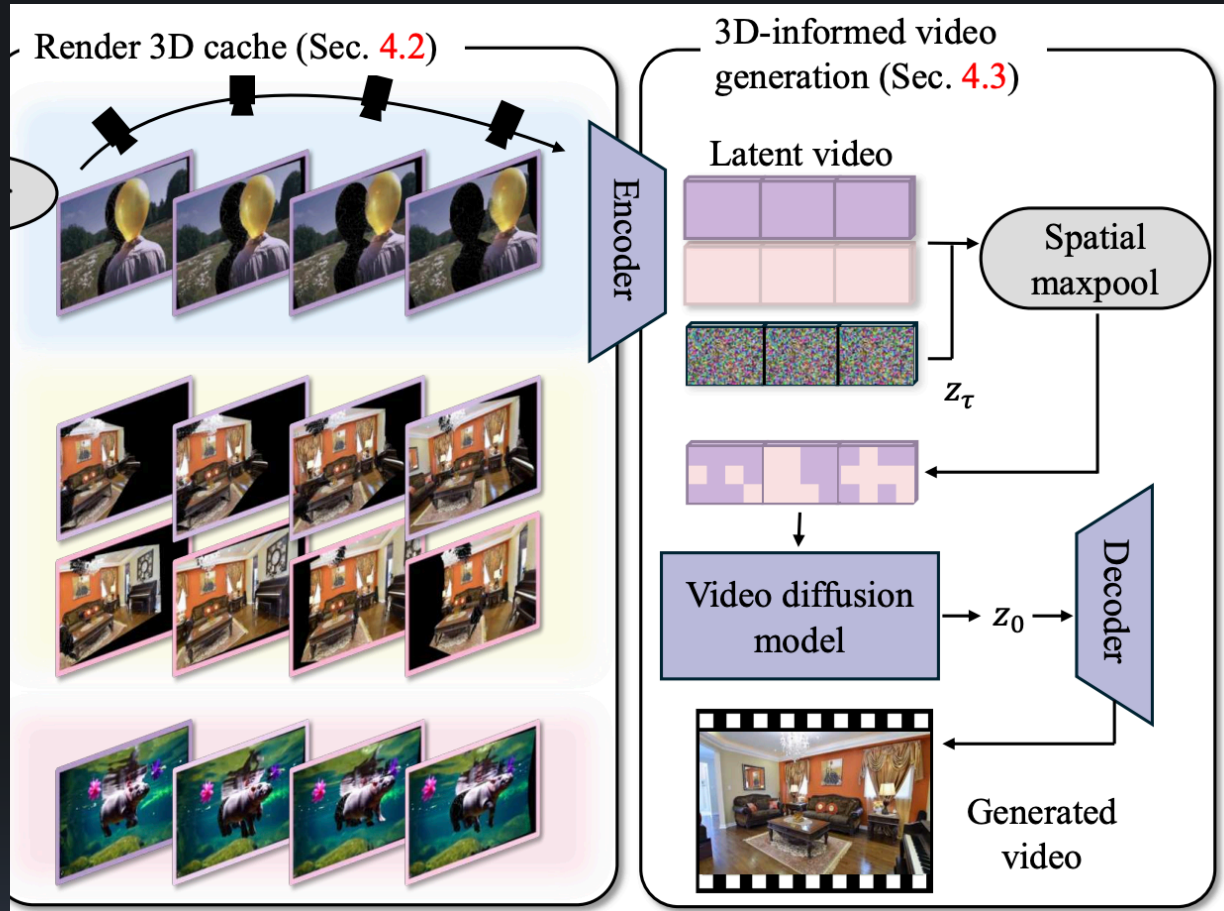
A **spatiotemporal 3D cache**

User defined **Camera Trajectory**

A sequence of new camera poses

- **Rendered Videos:** RGB images of the 3D cache as seen from the new camera poses.
- **Disocclusion Masks:** Binary masks that flag "empty" pixels not covered by the point cloud, identifying regions that the model needs to fill in.

Method Fusing and Injecting the 3D Cache



**Rendered
Videos**

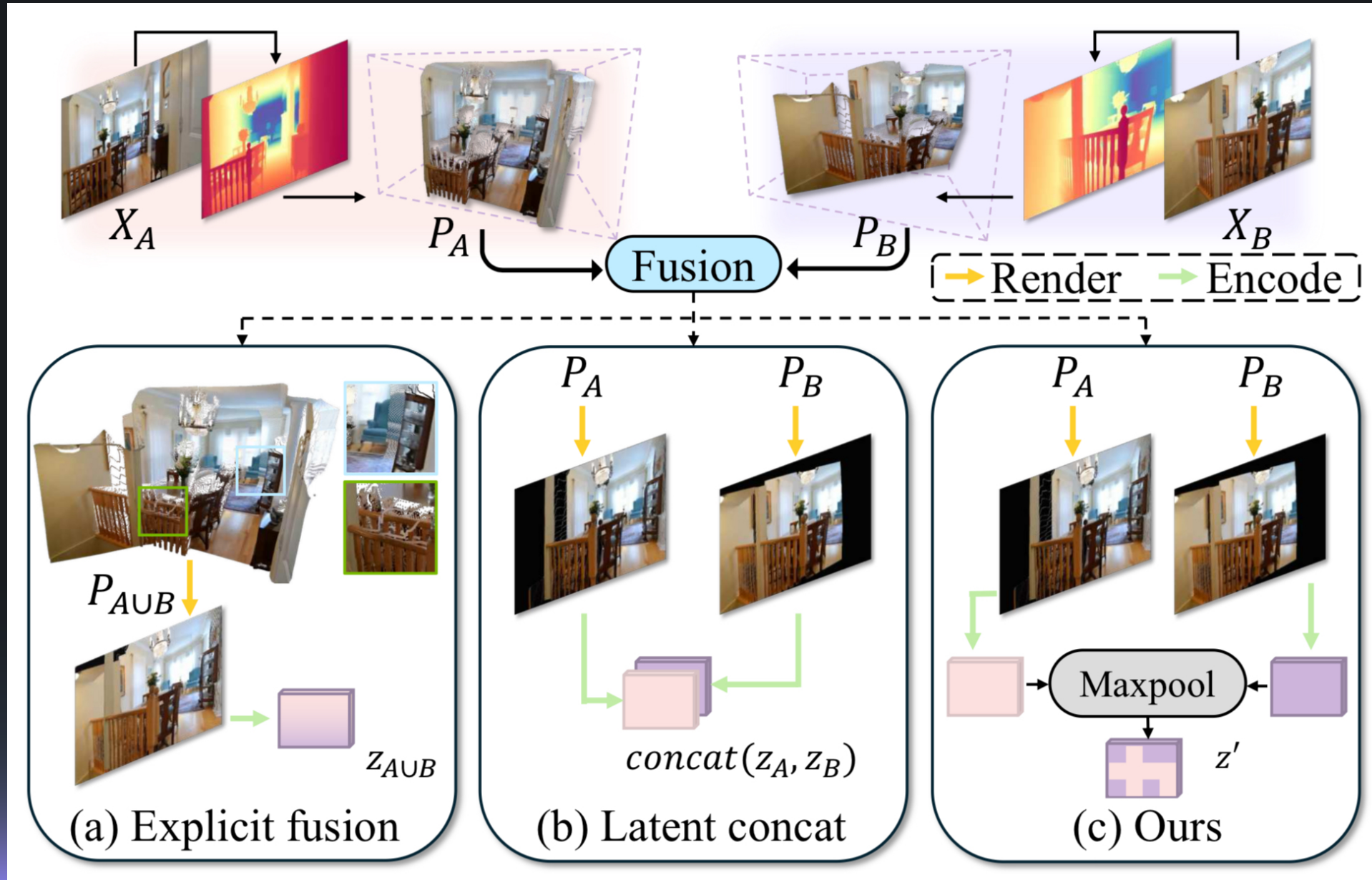
**Disocclusion
Masks**

First In-Layer encoding and
fusing multi-views 3D cache

**Noisy
Latents**

- **Fused Feature Map** : A single, unified latent representation that combines information from all viewpoints.
- **Conditioned Latent Video**: The final input fed into the subsequent layers of the video diffusion model to produce a consistent, high-quality video.

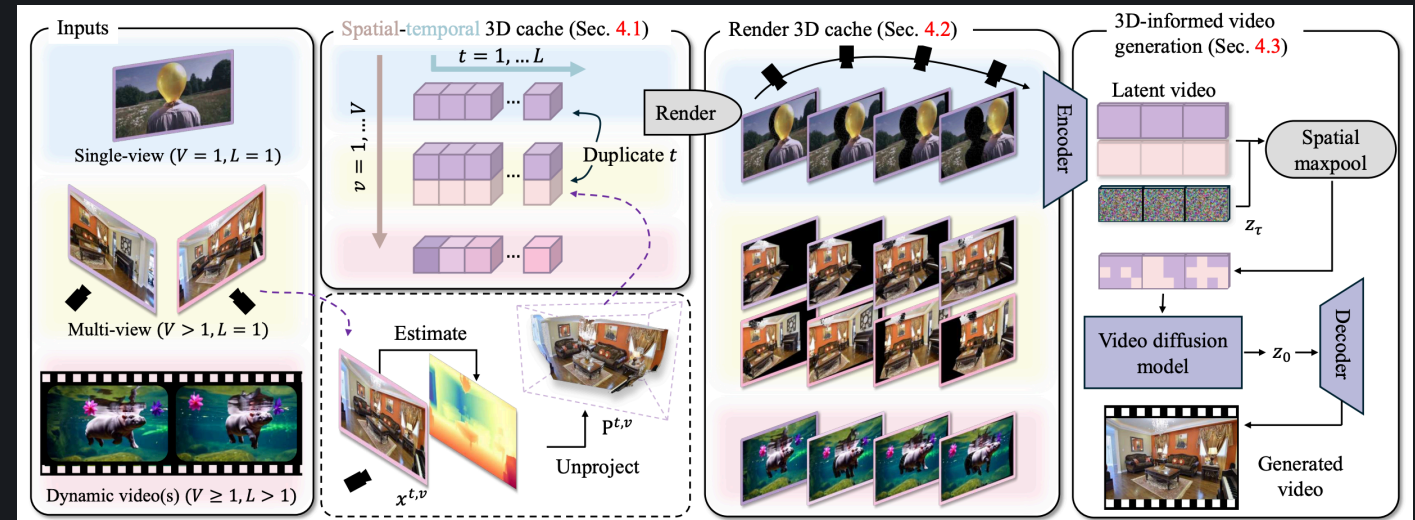
Method Fusing and Injecting the 3D Cache



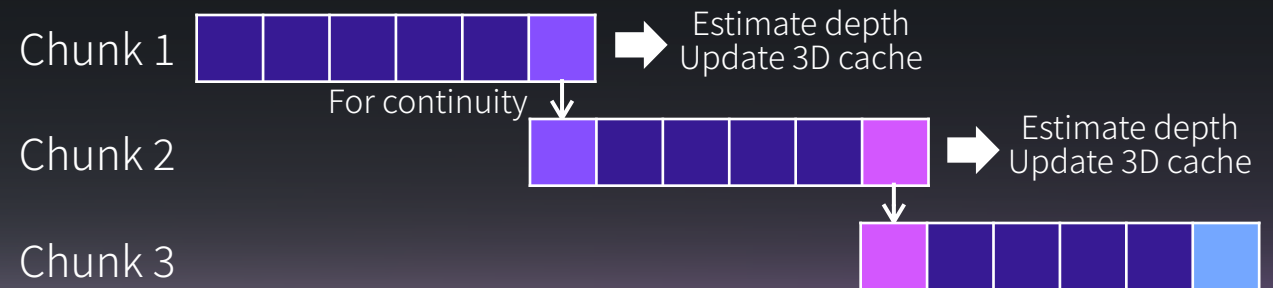
Training: Data Pairing

- Imperfect 3D cache renderings
- Ground Truth RGB video

Inference: 3D Cache Update



In order to process a long video (due to batch size limitation)



Validation **Single View to Video Generation**

This result proves that the 3D cache effectively prevents artifacts when the camera moves back and forth.

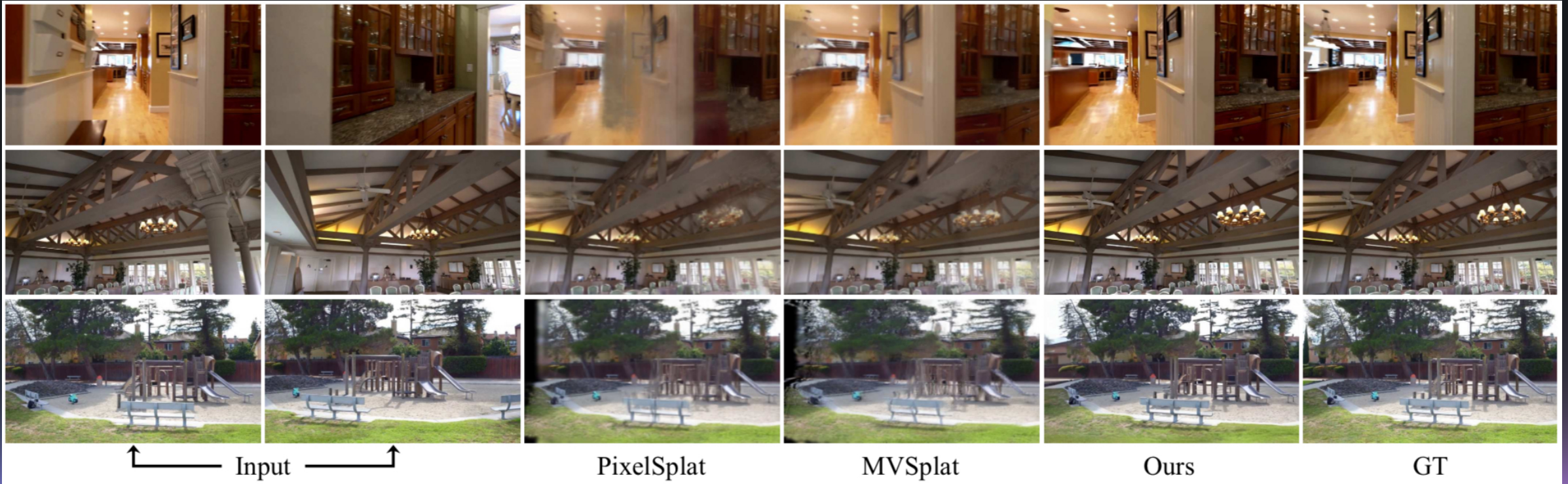
Method	Tanks-and-Temples [26]			RE10K [74]			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	TSED $\uparrow$
MotionCtrl [59]	13.46	0.46	0.42	13.60	0.59	0.46	0.1363
CameraCtrl [16]	15.88	0.55	0.29	18.40	0.72	0.25	0.8033
GenWarp [46]	16.04	0.50	0.39	15.50	0.61	0.40	0.0330
NVS-Solver [66]	16.95	0.59	0.27	16.90	0.69	0.30	0.7286
GEN3C	18.66	0.67	0.20	19.88	0.78	0.20	0.9143



Validation Two-Views Novel View Synthesis

This result presents superior performance in “filling in the gaps” (interpolation) and “seeing beyond” (extrapolation) between two provided images.

	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
T-&-T	PixelSplat [6]	21.34 / 17.45	0.70 / 0.65	0.42 / 0.46
	MVSplat [8]	20.90 / 16.08	0.70 / 0.63	0.39 / 0.44
	GEN3C	22.22 / 20.51	0.76 / 0.72	0.14 / 0.16
RE10K	PixelSplat [6]	19.74 / 16.95	0.75 / 0.70	0.40 / 0.44
	MVSplat [8]	21.40 / 15.51	0.78 / 0.69	0.30 / 0.37
	GEN3C	24.08 / 21.56	0.86 / 0.83	0.11 / 0.15



Validation **Monocular Dynamic Novel View Synthesis**

This result demonstrates better preservation of object dynamics and details under given novel camera poses.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
GCD [53]	19.37	0.67	0.48	150.64
GEN3C	19.41	0.63	0.29	98.58



Discussion

Reasons for Acceptance of This Paper

- They solve a long-standing limitation of novel view synthesis: only observed views are well represented
- Instead of using camera pose as a mere parameter that the model has difficulty connecting with pixel-level information, they used explicit 3D conditioning that can maintain 3D consistency effectively.
- Generalizable to different inputs: from single view, two views, and video to novel view video

Limitations of the paper

- Still has discrepancy with GT because the model “generates” occluded areas.

